
FINAL PROJECT REPORT

Smartphone Usage, Digital Addiction & Well-Being Analytics

An interactive Power BI analysis of screen time, addiction and well-being

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1. Introduction

1.1. Project Overview

Smartphones have become central to study, work and social life, but heavy and unplanned use is linked with lost focus, disturbed sleep and stress. Constant notifications, chat alerts and endless-scroll feeds pull users back to their phones again and again, and most users cannot see how much time they spend or which habits drive it.

This project, **Smartphone Usage, Digital Addiction & Well-Being Analytics**, uses Microsoft Power BI to analyse a dataset of **7,500 smartphone users**. The report combines daily and weekend screen time, social-media and gaming hours, notifications and app opens per day, sleep hours, stress level, work/study hours, academic-work impact, addiction level, addicted status, age group and gender. KPI cards, charts, slicers, a decomposition tree and a key-influencers visual let the reader explore how usage behaviour relates to addiction and well-being.

The Power BI file has five pages – **KPI, Visualization, Dashboard, Report** and **Report 2** – all built on one source table. SQL queries were also used to explore the same data (Section 10.1).

Total users	7,500	Avg. daily screen time	≈ 7.5 h	Addicted users 5,308 (70.8%)
Avg. weekend screen time	9.24 h	Avg. sleep	6.74 h	Avg. notifications / day 134.26

1.2. Objectives

- Analyse smartphone usage patterns and find the relationships between usage and digital well-being.
- Present key indicators – user count, daily screen time, sleep hours and weekend screen time – as clear KPIs.
- Examine social-media, gaming, notification and app-open behaviour to understand repeated phone checking.
- Compare addiction level and addicted status with sleep, social-media hours and weekend screen time.
- Compare usage across age groups and gender, and examine stress level and academic/work impact.
- Identify the factors most strongly associated with being addicted using the key-influencers analysis.
- Deliver an interactive dashboard and reports that students, counsellors and wellness advisors can use to plan healthier digital habits.

2. Project Initialization and Planning Phase

2.1. Define Problem Statement

Many people spend too much time on their smartphones, which creates a habit of screen addiction. They want to reduce screen time to focus on studies or work, but notifications, chat alerts and endless scrolling keep pulling them back. This project provides a simple analytics tool that tracks daily phone usage and finds the reasons behind the addiction, so people can see their habits through data, reduce screen time and gain peace of mind.

I am	A student or work-from-home professional who uses a smartphone too much every day.
I'm trying to	Complete daily tasks on time, focus deeply on work or studies, and get a good night's sleep.
But	Constant phone alerts, endless scrolling feeds and addictive apps keep breaking my concentration.
Because	I cannot see exactly how much time I waste on my phone, and I have no tools to block these distractions.
Which makes me feel	Anxious, tired, upset about wasting time, and helpless against my phone habits.

2.2. Project Proposal (Proposed Solution)

Objective	Analyse smartphone usage patterns with Power BI and identify relationships between smartphone use and digital well-being (screen time, social media, gaming, sleep, notifications, app opens, stress, work/study hours, academic impact and addiction measures).
Scope	KPI analysis, usage behaviour, addiction analysis, demographic comparison and well-being/impact analysis, using daily and weekend screen time, social-media and gaming hours, sleep, notifications, app opens, stress, work/study hours, academic impact, addiction level, Addicted Status, High Screen Time User, age group and gender.
Problem and impact	Usage patterns are spread across many measures. Bringing them together with sleep, stress and academic-impact data shows how usage differs across groups and which factors are connected with high usage or addicted status.
Approach	Power BI is used for data preparation, modelling, KPI creation and visualization with interactive charts, filters and summary measures, including a key-influencers analysis of Addicted Status.
Key features	KPI summary; screen-time, social-media and gaming analysis; weekend screen-time and sleep analysis; notifications and app opens; addiction level, Addicted Status and High Screen Time User; age-group and gender comparison; stress and academic/work impact; interactive dashboard and reports.
Resources	A system that runs Microsoft Power BI Desktop (Power Query and DAX); no external Python/R libraries; local storage for the PBIX file (about 0.28 MB) and data. Detailed hardware specifications were not specified.

2.3. Initial Project Planning

The backlog has nine user stories (21 story points) in three sprints: Sprint-1 = 6, Sprint-2 = 8, Sprint-3 = 7 points.

Sprint	Epic	USN	User Story / Task	Pts	Priority
Sprint-1	Data & KPI Analysis	USN-1	Prepare the smartphone-usage data and identify the main fields for analysis.	2	High
Sprint-1	Data & KPI Analysis	USN-2	Display KPIs: user count, daily screen time, sleep hours, weekend screen time.	2	High
Sprint-1	Usage Behaviour	USN-3	View social-media hours, gaming hours and notifications per day.	2	High
Sprint-2	Digital Addiction	USN-4	Analyse addiction level and addicted status using the Power BI visuals.	3	High
Sprint-2	Digital Addiction	USN-5	Compare addiction level with sleep hours, social-media hours and weekend screen time.	3	High
Sprint-2	Demographic Analysis	USN-6	Compare screen time and weekend screen time across age groups and gender.	2	Medium
Sprint-3	Well-Being Analysis	USN-7	Examine stress level together with app opens per day.	2	High
Sprint-3	Academic & Work Impact	USN-8	Analyse academic-work impact together with work/study hours.	2	High
Sprint-3	Dashboard & Insights	USN-9	View an integrated dashboard summarising addiction and well-being insights.	3	High

3. Data Collection and Preprocessing Phase

3.1. Data Collection Plan and Raw Data Sources Identified

The PBIX file has an embedded data model with a single source table, **smartphone** **smartphone_usage** (Power BI embedded model, about 0.28 MB, private project file). No external dataset URL or source-file location is exposed, so the original collection URL cannot be stated. **Fields:** demographic (age, Age Group, gender); usage (daily and weekend screen time, social media hours, gaming hours, app opens, notifications); well-being (sleep hours, stress level); academic/work (work study hours, academic work impact); addiction (addiction level, Addicted Status, addicted label, High Screen Time User); identifiers (user id, User Count, transaction id). **Limitation:** the original publisher, collection date, sampling method and permissions cannot be verified from the PBIX.

3.2. Data Quality Report

Data Source	Data Quality Issue	Severity	Resolution Plan
Telemetry Session Logging Logs	Missing hourly time features: only one raw timestamp was provided, blocking analysis by hour or time bracket.	High	Power Query calculated columns: Hour_of_Day, Day_of_Week, Time_Bracket.
App Analytics Classification Tables	Missing behaviour-group dimensions: app names had no macro categories for filtering.	High	Added conditional column App_Category to bundle the 7,500 rows into parent behavioural sets.
System Log History Backlog	Bloated footprint: extra system parameters (hash keys, thread IDs) increased file size.	Moderate	Used 'Choose Columns' to drop unneeded columns and keep core metrics.
Device Alert Tracking Profiles	Inconsistent duration format: usage time was loaded as text, breaking SUM/AVERAGE.	High	Calculated column casting text durations to whole integers (Session_Duration_Seconds); data types corrected.

3.3. Data Exploration and Preprocessing

Data overview	One flat table of 7,500 user records. Category totals for addiction level (2,874 + 2,434 + 1,373 + 819), age group (4,132 + 2,569 + 799) and stress level (2.56K + 2.50K + 2.44K) each add up to 7,500, so these fields are complete.
SQL exploration	Five queries profile the data: users and addicted users by gender; average screen time, sleep and share of high-stress users; missing-value check on age, screen time and addiction level; addicted label vs stress level; and user count, screen time and sleep by addiction level (code in Section 10.1).
Data cleaning	Missing-value check (SQL query 3); complete category totals in the report; unneeded system columns removed with Power Query "Choose Columns".
Data transformation	Derived fields: Age Group (Under 20, 20–29, 30–39); Addicted Status (Addicted 5,308 / Not Addicted 2,192, matching addicted label 1/0); High Screen Time User (High Usage 66.13% / Normal Usage 33.87%); User Count; plus the time and category columns in Section 3.2.
Data types	Numeric fields (screen time, social media, gaming, sleep, work/study hours, notifications, app opens, age) support average, sum and count; categorical fields (gender, stress level, addiction level, Age Group, Addicted Status, academic impact) drive axes, legends and slicers; text durations cast to integers.
Splitting / merging	No multi-valued text fields were found, so no splitting was needed; grouping columns were derived instead.
Data model	A single table with no relationships; every visual reads from it using average, sum or count aggregations.
Saved data	The cleaned data is stored in the embedded model of smartphone_usage.pbix; the SQL table smartphone_usage in the smartphone database holds the same records.

4. Data Visualization

4.1. Framing Business Questions

No.	Business Question	Finding
Q1	What is the average daily screen time by gender?	Male, Other and Female all average about 7.4–7.6 hours per day (Figure 1).
Q2	Which addiction level has the most users?	Moderate (2,874), then Severe (2,434), Mild (1,373) and None (819) (Figure 2).
Q3	What percentage of users are addicted and non-addicted?	About 5.31K (70.77%) have addicted label 1 and 2.19K (29.23%) label 0; the 5,308 matches the dashboard's Addicted count (Figure 3).
Q4	How does daily screen time vary across age groups?	30–39, 20–29 and Under 20 are all about 7.4–7.5 hours per day (Figure 4).
Q5	Is there a visible relationship between daily screen time and sleep hours?	Points spread across 3–12 h of screen time and 4.5–9 h of sleep with no clear linear pattern; a correlation test is needed (Figure 6).
Q6	How does social-media usage compare with gaming?	Social media averages about 3.3 h per day versus about 2.0 h for gaming (Figure 5).
Q7	Do gender groups differ substantially in screen time?	No – the bars in Figure 1 are closely aligned.
Q8	Do age groups differ substantially in screen time?	No – the bars in Figure 4 are nearly the same height.

4.2. Developing Visualizations

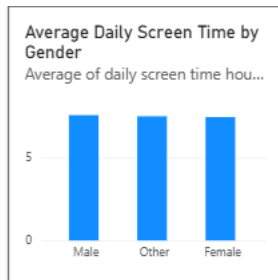


Figure 1. Daily screen time by gender

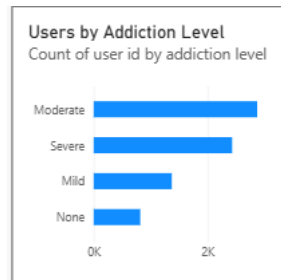


Figure 2. Users by addiction level

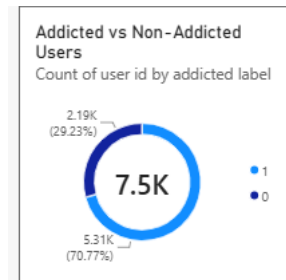


Figure 3. Addicted vs non-addicted

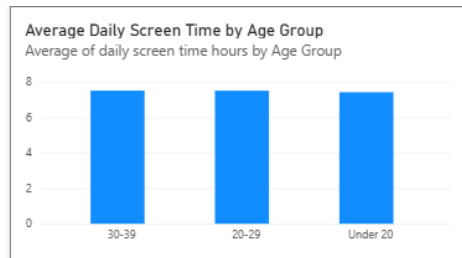


Figure 4. Daily screen time by age group

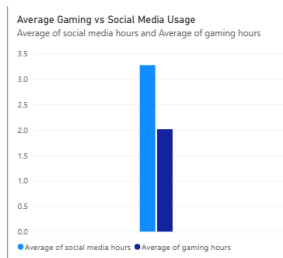


Figure 5. Social media vs gaming



Figure 6. Screen time vs sleep hours

5. Dashboard

5.1. Dashboard Design Screenshot with Insights

The **Digital Well-being & Addiction Insights Dashboard** shows how excessive phone use relates to daily routine, sleep and stress. It uses KPI cards on top, bar and pie charts below, distinct colours and an addiction-level slicer.

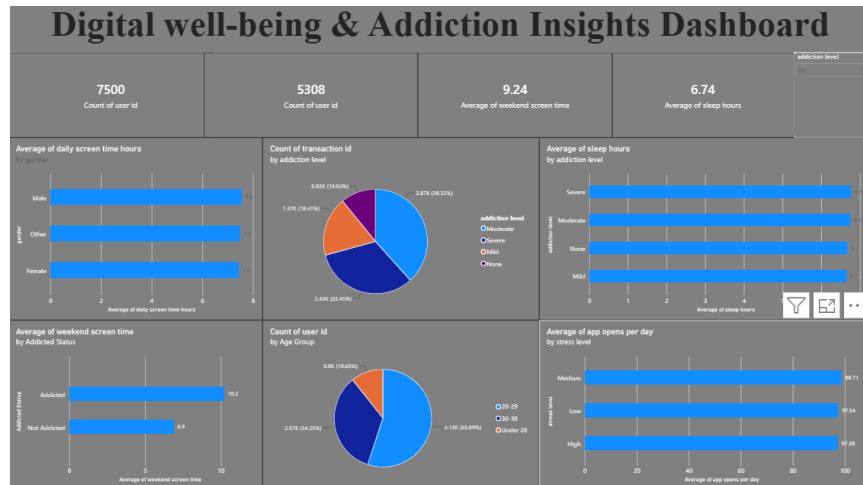


Figure 7. Digital Well-being & Addiction Insights Dashboard

- **Users and addiction:** 7,500 users; 5,308 (about 70.8%) are Addicted. Levels: Moderate 38.32% (2,874), Severe 32.45% (2,434), Mild 18.31% (1,373), None 10.92% (819).
- **Weekend and sleep:** average weekend screen time is 9.24 hours and sleep is 6.74 hours. Addicted users average 10.2 weekend hours versus 6.9 for Not Addicted users.
- **Gender and age:** daily screen time is almost identical across gender (7.6 male, 7.5 other, 7.4 female). Age 20–29 is 55.09% (4,132), 30–39 is 34.25% (2,569), Under 20 is 10.65% (799).
- **Stress:** average app opens per day are similar across stress levels (Medium 98.71, Low 97.54, High 97.28).

Faculty, counsellors and health advisors can use these insights to plan digital-detox sessions and guide students, especially young adults with long weekend screen time.

6. Report

6.1. Report Design Screenshot with Insights

Report 1 – Digital Addiction & Screen Time Behaviour Analysis

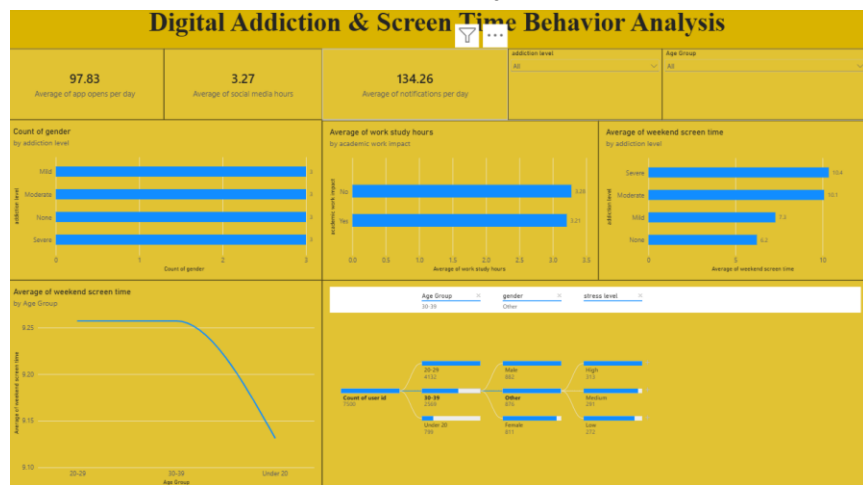


Figure 8. Report 1 – Digital Addiction & Screen Time Behaviour Analysis

- **Habits:** about 97.83 app opens, 3.27 hours of social media and 134.26 notifications per day, pointing to continuous checking.
- **Weekend trend:** weekend screen time rises with addiction: Severe 10.4 h, Moderate 10.1 h, Mild 7.3 h, None 6.2 h. Age groups 20–29 and 30–39 average 9.257 h; Under 20 average 9.132 h.
- **Work/study:** 3.28 h for users reporting no academic impact and 3.21 h for those reporting impact – nearly the same.

- **Decomposition tree:** the 30–39 group has 2,569 users (Male 882, Other 876, Female 811); in the Other branch stress is even: High 313, Medium 291, Low 272.

Report 2 – Smartphone Addiction & Digital Well-Being Insights

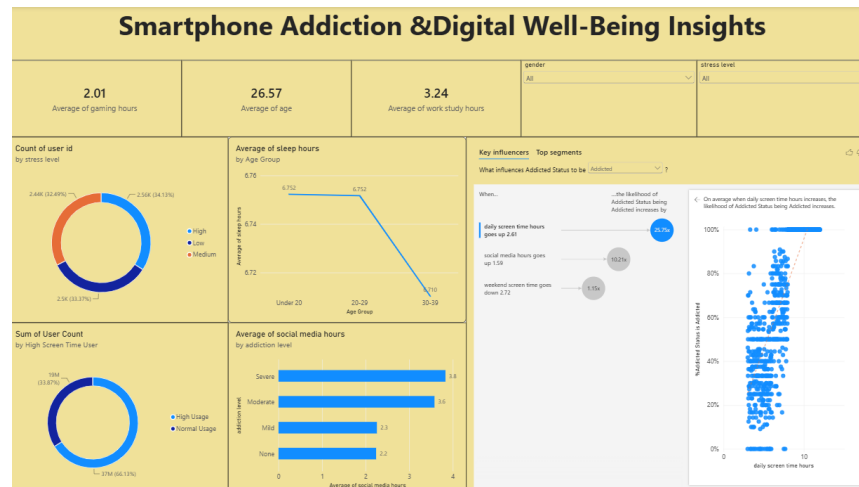


Figure 9. Report 2 – Smartphone Addiction & Digital Well-Being Insights

- **Profile:** average age 26.57 years (mostly students and young workers); gaming 2.01 h and work/study 3.24 h per day.
- **Stress and usage:** High 34.13% (2.56K), Low 33.37% (2.5K), Medium 32.49% (2.44K); 66.13% of users are High Usage and 33.87% Normal Usage.
- **Social media:** 3.8 h for Severe, 3.6 h Moderate, 2.3 h Mild and 2.2 h None. **Sleep:** 6.752 h for Under 20 and 20–29, 6.710 h for 30–39.
- **Key influencers of Addicted Status:** a 2.61-hour rise in daily screen time gives 25.75× higher likelihood of being Addicted; a 1.59-hour rise in social media gives 10.21×; weekend screen time shows 1.15× (when it goes down 2.72 h). These are associations, not proof of cause.

7. Performance Testing

This section is based on the report definition inside the PBIX. The model is one table of 7,500 rows and the file is about 0.28 MB, so the report is lightweight; Performance Analyzer load times were not part of the project files.

7.1. Utilization of Data Filters

The report uses **8 slicers** on four fields and **2 visual-level filters**; cross-filtering between visuals is also active.

Page	Slicers	Visual-level filters
KPI	–	User-count card: Addicted Status = Addicted
Visualization	Age Group, addiction level, gender	–
Dashboard	addiction level	User-count card: Addicted Status = Addicted (5,308)
Report	addiction level, Age Group	–
Report 2	gender, stress level	–

7.2. No. of Calculation Columns/Measures

Four derived fields are used: **Age Group** (Under 20, 20–29, 30–39), **Addicted Status** (Addicted / Not Addicted), **High Screen Time User** (High / Normal Usage) and **User Count**. No visual uses a named DAX measure; all values are aggregations (average, sum, count) of columns.

7.3. No. of Visualizations

The report has **5 pages** and **54 objects**: 43 data visuals (21 cards, 22 charts), 8 slicers and 3 text boxes.

Page	Cards	Charts	Slicers	Text boxes	Chart types
KPI	11	0	0	0	–
Visualization	0	6	3	0	3 column, 1 bar, 1 donut, 1 scatter

Page	Cards	Charts	Slicers	Text boxes	Chart types
Dashboard	4	6	1	1	4 bar, 2 pie
Report	3	5	2	1	3 bar, 1 line, 1 decomposition tree
Report 2	3	5	2	1	1 bar, 2 donut, 1 line, 1 key influencers
Total	21	22	8	3	

8. Conclusion/Observation

- **Addiction is widespread:** about 70.8% of users (5,308) are Addicted; Moderate (38.32%) and Severe (32.45%) are the largest levels.
- **Usage is heavy:** about 7.5 h of screen time per day, 9.24 h on weekends, about 98 app opens and 134 notifications daily; 66.13% are High Usage.
- **Weekends separate addicted users:** 10.2 h versus 6.9 h for Not Addicted users.
- **Social media matters:** 3.27 h versus 2.01 h for gaming, rising from 2.2 h (None) to 3.8 h (Severe).
- **Daily screen time is the strongest influencer:** +2.61 h gives 25.75× higher likelihood of being Addicted, followed by social media (10.21×).
- **Demographics differ little:** screen time is similar across gender and age; stress is split almost evenly; the screen-time vs sleep scatter shows no clear linear pattern.

These are associations in one dataset, not proof of cause, and the original data source could not be verified. Even so, reducing daily and weekend screen time, especially social media, is the most promising route to better focus, sleep and peace of mind.

9. Future Scope

- **Statistical validation:** correlations (screen time vs sleep and stress) and significance tests.
- **Predictive modelling:** classify addiction risk from usage behaviour, extending the key-influencers analysis.
- **Time-based analysis:** hourly and day-of-week trends and app-category breakdowns.
- **Real-time tracking and recommendations:** live usage data, limit alerts, digital-detox goals and focus schedules.
- **Report clean-up:** replace the “Count of gender by addiction level” chart (distinct count of 3 for every level) with a user count, and replace the User Count column (sums to millions in the donut) with a distinct-count measure.
- **Documentation and deployment:** record the original data source and sampling method, and publish to Power BI Service with scheduled refresh.

10. Appendix

10.1. Source Code

SQL queries used for exploratory analysis (smartphone_usage.sql). The Power BI report is in smartphone_usage.pbix.

```
use smartphone;
-- 1. Users and addicted users by gender
SELECT gender, COUNT(*), SUM(CASE WHEN addicted_label = 1 THEN 1 ELSE 0 END) AS addicted_count
FROM smartphone_usage GROUP BY gender;
-- 2. Overall averages and share of high-stress users
SELECT AVG(daily_screen_time_hours), AVG(sleep_hours), AVG(stress_level = 'High') FROM smartphone_usage;
-- 3. Missing-value check
SELECT * FROM smartphone_usage
WHERE age IS NULL OR daily_screen_time_hours IS NULL OR addiction_level IS NULL;
-- 4. Addicted label versus stress level
SELECT addicted_label, stress_level, COUNT(*) FROM smartphone_usage
GROUP BY addicted_label, stress_level ORDER BY addicted_label, stress_level;
-- 5. Profile of each addiction level
SELECT addiction_level, COUNT(*), AVG(daily_screen_time_hours), AVG(sleep_hours)
FROM smartphone_usage GROUP BY addiction_level;
```

10.2. GitHub & Project Demo Link

GitHub repository: <https://github.com/skasiya2005-oss/Smartphone-Usage-Digital-Addiction-and-Well-Being-Analytics.git>

Project demo video / dashboard link: https://drive.google.com/file/d/10UisU_DxRIrf_m-9jLrjJKmx3kST-9P0/view?usp=sharing